

Grading: Students answer exactly two problems. Each selected problem is worth 10 points, for a total of 20 points.

1. **Expected final answers:**

(a) Yes. (b) No.

Verification: The direction vectors satisfy

$$(2, 4, -2) = 2(1, 2, -1),$$

so the lines are parallel. They are distinct because $(0, 1, 4) - (1, 0, 2) = (-1, 1, 2)$ is not a scalar multiple of $(1, 2, -1)$. Thus they do not intersect.

Equivalent forms: “Parallel” for part (a) and “do not intersect” or “disjoint” for part (b) receive full credit.

Grading (10 points): 4 points for recognizing the proportional direction vectors, 3 points for showing that the lines are distinct, and 3 points for the two correct conclusions.

2. **Expected final answer:**

$$\vec{r}(t) = \left(5 \cos t, \frac{5}{2} \sin t \right), \quad \vec{v}(\pi/2) = \vec{r}'(\pi/2) = (-5, 0).$$

Verification: Substitution gives

$$(5 \cos t)^2 + 4 \left(\frac{5}{2} \sin t \right)^2 = 25(\cos^2 t + \sin^2 t) = 25,$$

and

$$\vec{v}(t) = \vec{r}'(t) = \left(-5 \sin t, \frac{5}{2} \cos t \right), \quad \vec{v}(\pi/2) = (-5, 0).$$

The parametrization starts at $(5, 0)$ and moves counterclockwise.

Equivalent forms: Algebraically identical notation is valid.

Any counterclockwise parametrization is also valid when paired with its derivative at $t = \pi/2$.

Grading (10 points): 6 points for the parametrization and 4 points for the derivative and its value at $t = \pi/2$.

3. **Expected final answer:**

$$\vec{r}(t) = (0, t, -3t), \quad t \in \mathbb{R}.$$

Verification: The plane normals are

$$\vec{n}_1 = (1, 3, 1), \quad \vec{n}_2 = (1, 6, 2), \quad \vec{n}_1 \times \vec{n}_2 = (0, -1, 3).$$

Setting $y = 0$ in both plane equations gives the point $P = (0, 0, 0)$. Thus $\vec{r}(t) = t(0, -1, 3)$ parametrizes the intersection; replacing t by $-t$ gives the displayed answer.

Equivalent forms: Any affine parametrization of the same line $x = 0, z = -3y$ receives full credit. In particular, any nonzero scalar multiple of the direction vector $(0, 1, -3)$ is valid.

Grading (10 points): 4 points for a correct cross product, 2 points for a point on both planes, and 4 points for a complete parametric equation.

4. **Expected final answer:**

$$t = 2, \quad \vec{r}(2) = (2, 4, 4).$$

Verification: At the film, $t + t^2 + 2t = 10$, so $t^2 + 3t - 10 = (t + 5)(t - 2) = 0$. Because $t \geq 0$, the only allowed value is $t = 2$, and $\vec{r}(2) = (2, 4, 4)$. The point satisfies $2 + 4 + 4 = 10$.

Equivalent forms: The time must equal 2; the point may be written as an ordered triple or with labeled coordinates.

Grading (10 points): 4 points for the plane-intersection equation, 3 points for the allowed parameter value, and 3 points for the point of intersection.